FUNKCIA VIACERÝCH PREMENNÝCH

$f(x) = x^2$

$f(x,y) = x^2 + 2xy$

$f(x,y,z) = x^2 + 2xy + z^3$

$x^2 + y^2 + z^2 = 1$ ✗

$f(x,y,z)$

$z = \sqrt{1 - x^2 - y^2}$ ✓

DEFINIČNÝ OBOR

↳ množina všetkých hodnôt, ktoré môže daná funkcia nadobúdať, aby mala riešenie

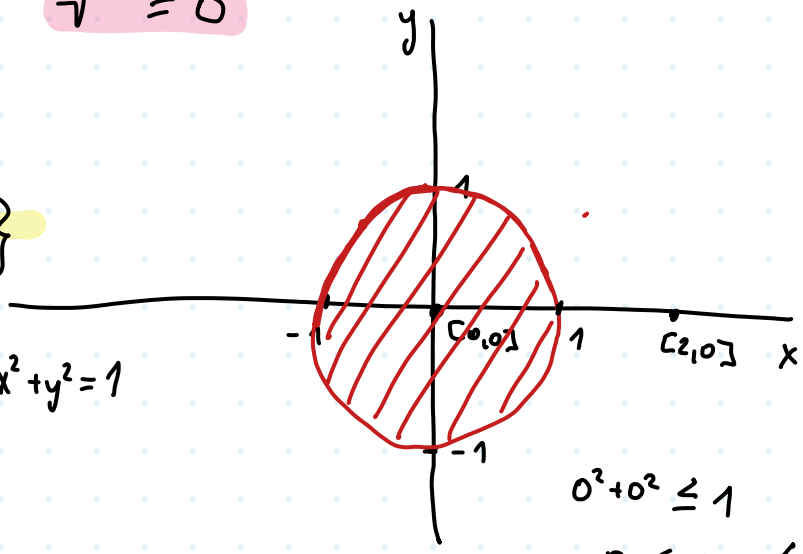
Príklad $f(x,y) = \sqrt{1 - x^2 - y^2}$

$\sqrt{\ } \geq 0$

Podmienka: $1 - x^2 - y^2 \geq 0$

$1 \geq x^2 + y^2$

$D(f) = \{ [x,y] \in R^2 \mid x^2 + y^2 \leq 1 \}$



$x^2 + y^2 = 1$

$0^2 + 0^2 \leq 1$

$0 \leq 1$ ✓

$2^2 + 0^2 \leq 1$

$4 \leq 1$ ✗

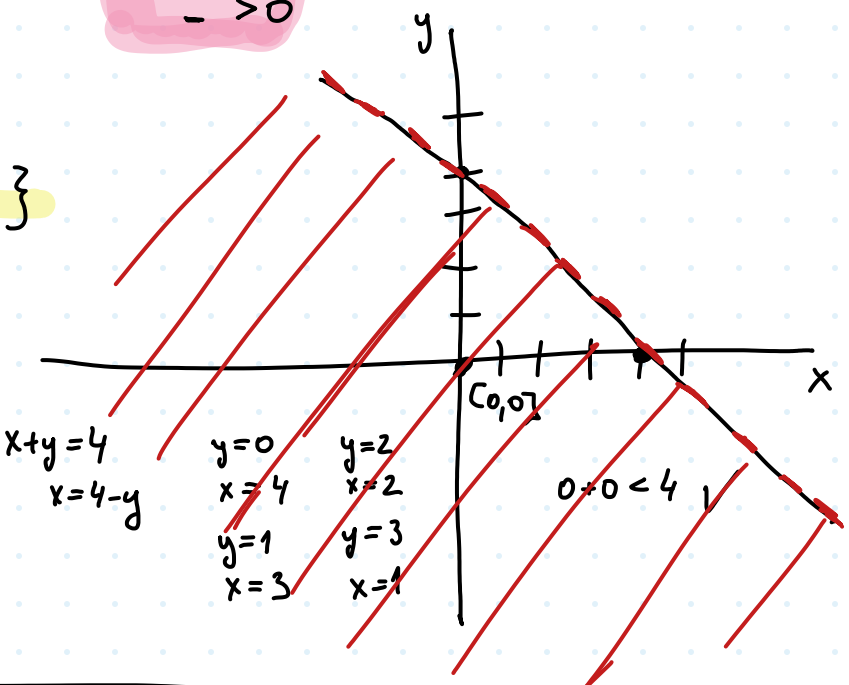
Przykład $f(x,y) = \ln(4-x-y)$

$\ln(_)$ $_ > 0$

Podmienka: $4-x-y > 0$

$4 > x+y$

$D(f) = \{[x,y] \in R^2 \mid x+y < 4\}$

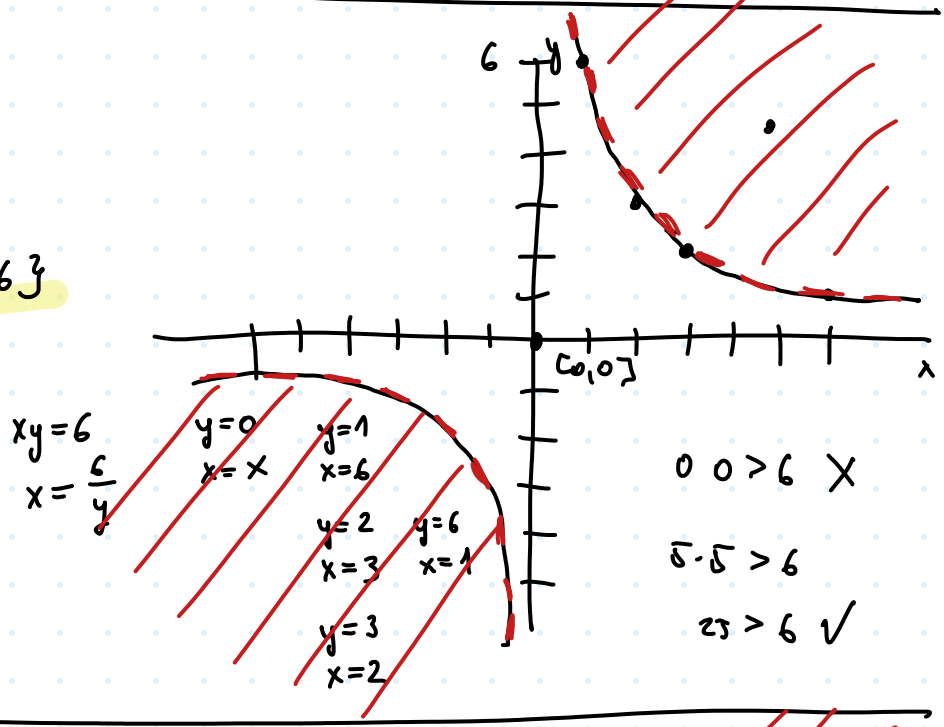


Przykład $f(x,y) = \ln(xy-6)$

Podmienka: $xy-6 > 0$

$xy > 6$

$D(f) = \{[x,y] \in R^2 \mid xy > 6\}$

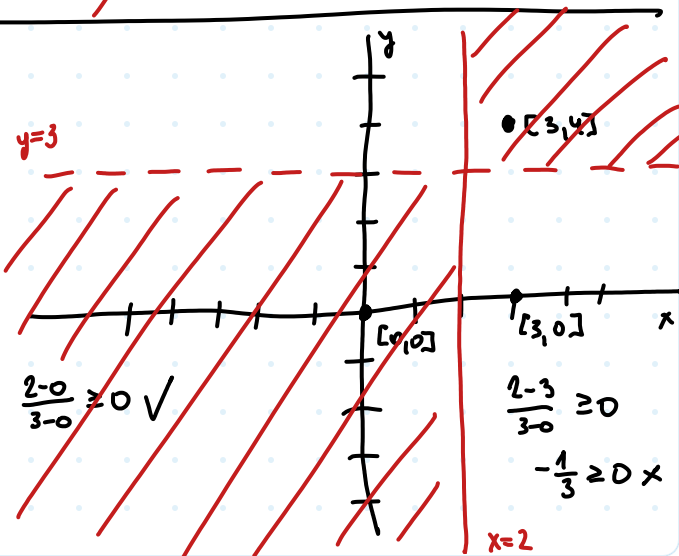


Przykład $f(x,y) = e^{\sqrt{\frac{2-x}{3-y}}}$

Podmienka: $\frac{2-x}{3-y} \geq 0$ $\quad 3-y \neq 0$

$y \neq 3$

$D(f) = \{[x,y] \in R^2 \mid \frac{2-x}{3-y} \geq 0,\ y \neq 3\}$

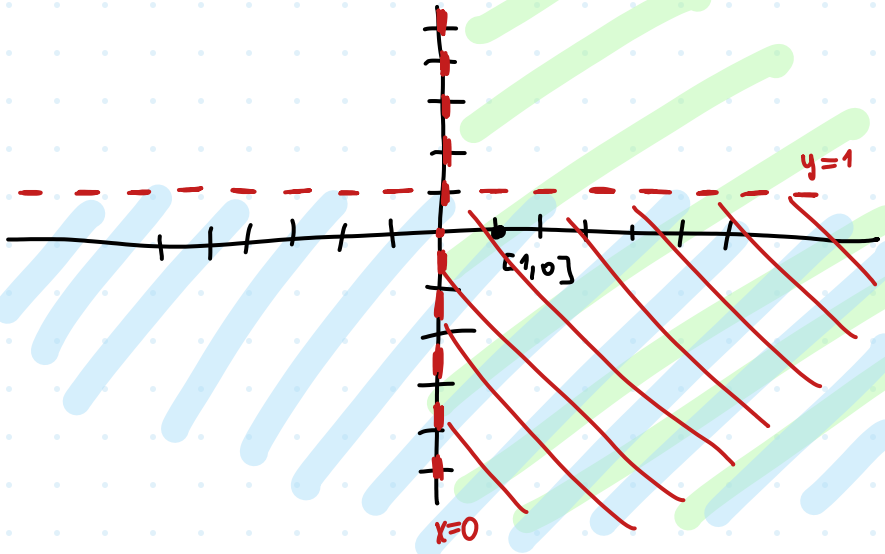


Príklad $f(x,y) = \dfrac{\ln(x)-y}{\sqrt{1-y}}$

Podmienka: $x > 0 \;\wedge\; \sqrt{1-y} \neq 0 \;\wedge\; 1-y \geq 0$

$1-y > 0 \qquad 1 > y$

$$D(f) = \{ [x,y] \in \mathbb{R}^2 \mid x > 0 \;\wedge\; y < 1 \}$$

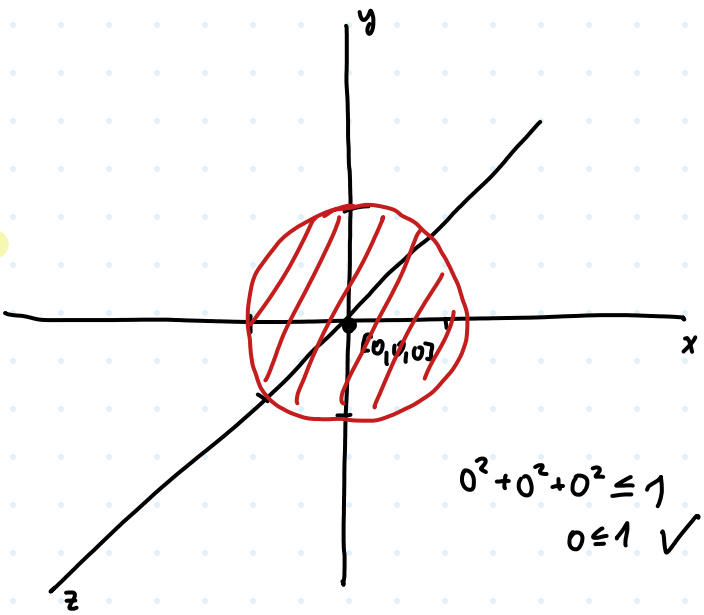


Príklad $f(x,y,z) = \sqrt{1-x^2-y^2-z^2}$

Podmienka: $1-x^2-y^2-z^2 \geq 0$

$x^2+y^2+z^2 \leq 1$

$$D(f) = \{ [x,y,z] \in \mathbb{R}^3 \mid x^2+y^2+z^2 \leq 1 \}$$



$0^2+0^2+0^2 \leq 1$

$0 \leq 1$ ✓

PARCIALNE DERIVACIE

-2-

L.p.	Funkcja	Pochodna
1	a	0
2	x	1
3	x^n	$n \cdot x^{n-1}$
4	$a \cdot f(x)$	$a \cdot f'(x)$
5	$\sin(x)$	$\cos(x)$
6	$\cos(x)$	$-\sin(x)$
7	$\frac{1}{x}$	$-\frac{1}{x^2}$
8	$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$
9	e^x	e^x
10	a^x	$a^x \cdot \ln(a)$
11	$\log_a(x)$	$\frac{1}{x \ln(a)}$
12	$\ln(x)$	$\frac{1}{x}$
13	$\operatorname{tg}(x)$	$\frac{1}{\cos^2(x)}$
14	$\operatorname{ctg}(x)$	$-\frac{1}{\sin^2(x)}$
15	$\operatorname{arc}\sin(x)$	$\frac{1}{\sqrt{1-x^2}}$
16	$\operatorname{arc}\cos(x)$	$-\frac{1}{\sqrt{1-x^2}}$
17	$\operatorname{arctg}(x)$	$\frac{1}{1+x^2}$
18	$\operatorname{arcctg}(x)$	$-\frac{1}{1+x^2}$
19	$f(x)+g(x)$	$f'(x)+g'(x)$
20	$f(x)-g(x)$	$f'(x)-g'(x)$
21	$f(x) \cdot g(x)$	$f'(x) \cdot g(x) + f(x) \cdot g'(x)$
22	$\frac{f(x)}{g(x)}$	$\frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{g^2(x)}$

$5' = 0$

$x = 1$

$x^2 = 2 \cdot x$ $\quad x^3 = 3x^2$

$5^x = 5^x \cdot \ln 5$

$\ln 5' = \frac{1}{5}$

$x^2 + x^3 = 2x + 3x^2$

$x^2 - x^3 = 2x - 3x^2$

$x^2 \cdot x^3 = 2x \cdot x^3 + x^2 \cdot 3x^2$

$\frac{x^2}{x^3} = \frac{2x \cdot x^3 - x^2 \cdot 3x^2}{(x^3)^2}$

$f(x,y) = x^2 + 2xy - \ln y$

Parciálna derivácia podľa x · $\frac{\partial f}{\partial x} = f'_x \Rightarrow y = \text{konštanta}$

Parciálna derivácia podľa y · $\frac{\partial f}{\partial y} = f'_y \Rightarrow x = \text{konštanta}$

Príklad $f(x,y) = 3x + x^2y^2 + 2x^3y$

y = konštanta

$f'_x = 3 \cdot 1 + \left(2x \cdot y^2 + x^2 \cdot 0\right) + 2 \cdot 3x^2 \cdot y = 3 + 2xy^2 + 6x^2y$

$2xy^2$

x = konštanta

$f'_y = 0 + x^2 \cdot 2y + 2x^3 \cdot 1 = 2x^2y + 2x^3$

Príklad $f(x,y) = 4x^3y + 3x^2y^2 - y^3$

y = konštanta

$f'_x = 4 \cdot 3x^2 \cdot y + 3 \cdot 2x \cdot y^2 - 0 = 12x^2y + 6xy^2$

x = konštanta

$f'_y = 4x^3 \cdot 1 + 3x^2 \cdot 2y - 3y^2 = 4x^3 + 6x^2y - 3y^2$

Príklad $f(x,y) = y \sin x + \cos(2x-y)$

$\square = (2x-y)' = 2 - 0$

y = konštanta

$f'_x = y \cdot \cos x + \left(-\sin(2x-y) \cdot (2-0)\right) = y\cos x - 2\sin(2x-y)$

x = konštanta

$f'_y = 1 \cdot \sin x + \left(-\sin(2x-y) \cdot (0-1)\right) = \sin x + \sin(2x-y)$

Příklad $f(x,y) = e^{\frac{x}{y}} + x^y$

x^y
x^2

y=konstanta

$f'_x = e^{\frac{x}{y}} \cdot \square + y \cdot x^{y-1}$

$\square = \left(\frac{x}{y}\right)' = \frac{1 \cdot y - x \cdot 0}{y^2} = \frac{y}{y^2} = \frac{1}{y}$

$= e^{\frac{x}{y}} \cdot \frac{1}{y} + y \cdot x^{y-1}$

x=konstanta

$f'_y = e^{\frac{x}{y}} \cdot \square + x^y \cdot \ln x$

x^y
2^y

$\square = \left(\frac{x}{y}\right)' = \frac{0 \cdot y - x \cdot 1}{y^2} = \frac{-x}{y^2}$

$= e^{\frac{x}{y}} \cdot \frac{-x}{y^2} + x^y \cdot \ln x$

Příklad $f(x,y) = \frac{y}{\sqrt{x}}$

$(\sqrt{x})' = \left(x^{\frac{1}{2}}\right)' = \frac{1}{2} \cdot x^{\frac{1}{2}-1} = \frac{1}{2} \cdot x^{-\frac{1}{2}}$

$\sqrt[3]{x} = x^{\frac{1}{3}}$

$x^{-\frac{1}{2}}$

$\frac{1}{x^{\frac{1}{2}}}$

y=konstanta

$f'_x = \frac{0 \cdot \sqrt{x} - y \cdot \square}{(\sqrt{x})^2} = \frac{-y \cdot \frac{1}{2} \cdot x^{-\frac{1}{2}}}{x} = \frac{-y \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{x}}}{x}$

$= \frac{\frac{-y}{2\sqrt{x}}}{\frac{x}{1}} = \frac{-y}{2x\sqrt{x}}$

x=konstanta

$f'_y = \frac{1 \cdot \sqrt{x} - y \cdot 0}{(\sqrt{x})^2} = \frac{\sqrt{x}}{x}$

Príklad $f(x,y) = \ln\left(\frac{x+y}{x-y}\right)$ $\qquad \ln x' = \frac{1}{x}$

y = konštanta

$$f'_x = \frac{1}{\left(\frac{x+y}{x-y}\right)} \cdot \frac{(1+0)\cdot(x-y) - (x+y)\cdot(1-0)}{(x-y)^2} = \frac{x-y}{x+y} \cdot \frac{(x-y)-(x+y)}{(x-y)^2}$$

x = konštanta

$$f'_y = \frac{1}{\left(\frac{x+y}{x-y}\right)} \cdot \frac{(0+1)(x-y) - (x+y)(0-1)}{(x-y)^2} = \frac{x-y}{x+y} \cdot \frac{(x-y)+(x+y)}{(x-y)^2}$$

Príklad $f(x,y) = \operatorname{arctan}\left(\frac{x-y}{1+xy}\right)$ $\qquad \operatorname{arctan} x' = \frac{1}{1+x^2}$

y = konštanta

$$f'_x = \frac{1}{1+\left(\frac{x-y}{1+xy}\right)^2} \cdot \frac{(1-0)(1+xy) - (x-y)(0+y)}{(1+xy)^2} = \frac{1}{1+\left(\frac{x-y}{1+xy}\right)^2} \cdot \frac{(1+xy) - y(x-y)}{(1+xy)^2}$$

$$f'_y = \frac{1}{1+\left(\frac{x-y}{1+xy}\right)^2} \cdot \frac{(0-1)(1+xy) - (x-y)(0+x)}{(1+xy)^2} = \ldots$$

Príklad $f(x,y,z) = x^2y^2 + xyz - z^3$

y, z = konštanty

$$f'_x = 2xy^2 + 1\cdot y\cdot z - 0 = 2xy^2 + yz$$

x, z = konštanty

$$f'_y = x^2 2y + x\,1\cdot z - 0 = 2x^2y + xz$$

x, y = konštanty

$$f'_z = 0 + xy\cdot 1 - 3z^2 = xy - 3z^2$$